

Exercise 04

Arham Lodha

March 12, 2025

Problem 1a.

Solution. 1. False

2. False

3. True

4. True

5. True

■

Problem 1b.

Solution. 1. True

2. False

3. True

4. False

5. True

■

Problem 2a.

Proof. $\forall x \in \mathbb{Z}$ and let $\alpha = \frac{1}{2}$. We want to show that $E_x(V_x) = \infty$. Note $p_{x,x}^{2n} = 2^{-2n} \binom{2n}{n} \sim \frac{c}{\sqrt{n}}$ as $n \rightarrow \infty$ as given.

$$\begin{aligned} E_x(V_x) &= \sum_{n=1}^{\infty} p_{0,0}^{(n)} \\ &= \sum_{n=1}^{\infty} p_{0,0}^{(2n)} \\ &= \sum_{n=1}^{\infty} \binom{2n}{n} \left(\frac{1}{2}\right)^{2n} \\ &\geq \sum_{n=1}^{\infty} \frac{c}{2\sqrt{n}} = \infty \end{aligned}$$

Thus $E_x(V_x) = \infty$. Thus x is recurrent. $\square$

Problem 2b.

Proof. Let $\{Y_i\}_{i \geq 1}$ be a sequence of random variables where $\mathbb{P}(Y_i = 1) = \alpha$ and $\mathbb{P}(Y_i = -1) = 1 - \alpha$. Thus $X_n = \sum_{m=1}^n Y_m$.

By the Strong Law of Large Numbers, $\frac{X_n}{n} \rightarrow E[Y_m] = \alpha - 1 + \alpha = 2\alpha - 1$ almost surely. For $\alpha < \frac{1}{2}$: $\frac{X_n}{n}$ approaches a value less than 0. Thus $X_n \rightarrow -\infty$. Similarly for $\alpha > \frac{1}{2}$: $\frac{X_n}{n}$ approaches a value greater than 0. Thus $X_n \rightarrow \infty$. Thus in either case: x is a transient state. $\square$

Problem 2c.

Proof. Let $Y_n = |X_n|$.

1. For $\alpha = \frac{1}{2}$, X_n is recurrent at 0. Thus Y_n is recurrent at 0. Let Z_n be such a reflected walk.
2. For $\alpha < \frac{1}{2}$: Couple Y_n and Z_n in the following way:
 - (a) If $Y_n = 0$: Move right. Otherwise
 - (b) If Z_n moves left then Y_n moves left.
 - (c) If Z_n moves right then Y_n we pick from the leftover portion of Y 's distribution for the "right step." Since $\alpha < \frac{1}{2}$, this leftover portion is enough to ensure we can couple them consistently.

Let T_k be the k th visiting time of Z on 0. We want to show for all $k \geq 1$, $\exists n \in [T_k, T_{k+1}] \ni Y_n = 0$. Let l_z, l_y, r_z, r_y , be the left steps of Z and Y and the right steps of Z and Y .

Assume the contrary $\forall n \in [T_k, T_{k+1}]$, $Y_n \neq 0$. In the interval, $[T_k, T_{k+1}]$, $l_z = r_z$. Also note because of the coupling, $l_y \geq l_z$. Since $l_z + r_z = T_{k+1} - T_k = l_y + r_y$, at $Y_{T_{k+1}} \leq Z_{T_{k+1}} = 0$. However $Y_{T_{k+1}} \geq 0 \implies Y_{T_{k+1}} = 0$, which is a contradiction.

Thus $\exists n \in [T_k, T_{k+1}] \ni Y_n = 0$. Thus Y visits 0 infinitely often since Z visits 0 infinitely often. Thus 0 is recurrent for Y . $\square$

Problem 3.

Proof. $\square$

Let Y_n be a SRW on a d -regular tree. $\forall x \in V$. Let $D_n := |Y_n - x|$ or distance from Y_n to x . D_n is a biased reflected random walk where $\alpha = \frac{d-1}{d}$. Thus D_n isn't transient which means Y_n isn't transient on x .

Problem 4a.

Proof. $\forall (x, y) \in \mathbb{Z}^2 \setminus \{(0, 0)\}$, $\mathbb{P}_{(x, y)}(H_{(0, 0)} = |x| + |y|) = 1$.

Thus,

$$\begin{aligned} \mathbb{P}_{(0, 0)}(H_{(0, 0)} < \infty) &= \sum_{(x, y) \in \mathbb{Z}^2} \mathbb{P}_{(0, 0)}(X_1 = (x, y)) \mathbb{P}_{(x, y)}(H_{(0, 0)} < \infty \mid X_1 = (x, y)) \\ &= \mu(0, 0) + \sum_{(x, y) \in \mathbb{Z}^2 \setminus \{(0, 0)\}} \mu(x, y) \mathbb{P}_{(x, y)}(H_{(0, 0)} < \infty) \\ &= \mu(0, 0) + \sum_{(x, y) \in \mathbb{Z}^2 \setminus \{(0, 0)\}} \mu(x, y) = 1 \end{aligned}$$

Thus $(0, 0)$ is recurrent. □

Problem 4b.

Proof.

$$\begin{aligned} E_0(H_0) &= \mathbb{P}_{(0, 0)}(X_1 = (0, 0)) + \sum_{(x, y) \in \mathbb{Z}^2 \setminus \{0, 0\}} \mathbb{P}_{(0, 0)}(X_1 = (x, y)) (E_{(x, y)}(H_0) + 1) \\ &= \mu(0, 0) + \sum_{(x, y) \in \mathbb{Z}^2 \setminus \{(0, 0)\}} \mu(x, y)(|x| + |y| + 1) \\ &= 1 + \sum_{(x, y) \in \mathbb{Z}^2 \setminus \{(0, 0)\}} \mu(x, y)(|x| + |y|) \\ &= 1 + E_\mu[|x| + |y|] \end{aligned}$$

$(0, 0)$ is positive recurrent if $E_\mu[|x| + |y|]$ is finite and null recurrent otherwise. □

Problem 4c.

Proof. 1. Let $Y_n = \|X_n\|_\infty$.

$$\mathbb{P}_0(Y_2 = 2 \mid Y_1 = \|(2, 2)\|_\infty = 2) = 1$$

and

$$\mathbb{P}(Y_2 = 2 \mid Y_1 = 2) = \frac{1}{2}$$

Thus Y_n is not a markov processs since $\mathbb{P}(Y_2 = 2 \mid Y_1 = 2) \neq \mathbb{P}_0(Y_2 = 2 \mid Y_1 = \|(2, 2)\|_\infty = 2)$

2. Let $Y_n = \|X_n\|_1$. From any state with L^1 norm k for $k \neq 0$, the next norm is $k - 1$ deterministically. For $k = 0$,

$$p_{k, n} = \sum_{|x| + |y| = n} \mu(x, y).$$

The transitions depend only on the current norm, satisfying the Markov property

3. The projection onto the x-coordinate is not Markov. The transition of the x-coordinate depends on the underlying y-coordinate, which is not captured by the projection. Different underlying states with the same x-coordinate can lead to different future x-coordinates, violating the Markov property □